01 Discrete Structures 200Q

Topic 1 Discrete Structures - 200 MCQs

Syllabus: Set Theory (Sets, Relations, Functions, Pigeonhole, Inclusion-Exclusion, Equivalence/Partial Orderings, Counting, Probability, Information measures), Graph (walks, paths, trails, connected/regular/bipartite, cycles/circuits, trees/rooted/spanning, eccentricity/radius/diameter/centre, Hamiltonian/Eulerian, Planar), Groups (Finite fields, Error correcting/detecting codes). Distribution: ~70 set theory/counting, ~100 graph/tree, ~30 groups/coding.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

---

Q1. In a survey, |A|=52, |B|=27, |A ? B|=22. What is |A ? B|?

A. 59

Q2. In a survey, |A|=57, |B|=47, |A ? B|=18. What is |A ? B|?

A. 86

Q3. In a survey, |A|=59, |B|=28, |A ? B|=15. What is |A ? B|?

A. 71

Q4. In a survey, |A|=27, |B|=25, |A ? B|=7. What is |A ? B|?

A. 45

Q5. In a survey, |A|=27, |B|=45, |A ? B|=15. What is |A ? B|?

A. 57

Q6. In a survey, |A|=42, |B|=28, |A ? B|=7. What is |A ? B|?

A. 63

Q7. In a survey, |A|=28, |B|=22, |A ? B|=15. What is |A ? B|?

A. 35

Q8. In a survey, |A|=29, |B|=42, |A ? B|=7. What is |A ? B|?

A. 64

Q9. In a survey, |A|=15, |B|=44, |A ? B|=8. What is |A ? B|?

A. 50

Q10. In a survey, |A|=38, |B|=60, |A ? B|=28. What is |A ? B|?

A. 69

Q11. For sets A,B,C: |A|=21, |B|=33, |C|=25, |A?B|=9, |B?C|=6, |A?C|=7, |A?B?C|=5. Find |A?B?C|.

A. 62

Q12. For sets A,B,C: |A|=25, |B|=35, |C|=36, |A?B|=9, |B?C|=7, |A?C|=5, |A?B?C|=2. Find |A?B?C|.

A. 75

Q13. For sets A,B,C: |A|=37, |B|=34, |C|=23, |A?B|=12, |B?C|=7, |A?C|=7, |A?B?C|=3. Find |A?B?C|.

A. 72

Q14. For sets A,B,C: |A|=24, |B|=29, |C|=30, |A?B|=11, |B?C|=6, |A?C|=5, |A?B?C|=6. Find |A?B?C|.

A. 70

Q15. For sets A,B,C: |A|=26, |B|=28, |C|=30, |A?B|=6, |B?C|=9, |A?C|=7, |A?B?C|=4. Find |A?B?C|.

A. 64

Q16. For sets A,B,C: |A|=23, |B|=36, |C|=30, |A?B|=8, |B?C|=6, |A?C|=9, |A?B?C|=3. Find |A?B?C|.

A. 67

Q17. For sets A,B,C: |A|=39, |B|=34, |C|=40, |A?B|=5, |B?C|=10, |A?C|=8, |A?B?C|=4. Find |A?B?C|.

A. 95

Q18. For sets A,B,C: |A|=37, |B|=38, |C|=37, |A?B|=10, |B?C|=10, |A?C|=5, |A?B?C|=3. Find |A?B?C|.

A. 90

Q19. For sets A,B,C: |A|=32, |B|=32, |C|=40, |A?B|=5, |B?C|=8, |A?C|=6, |A?B?C|=3. Find |A?B?C|.

A. 91

Q20. For sets A,B,C: |A|=24, |B|=30, |C|=23, |A?B|=10, |B?C|=5, |A?C|=12, |A?B?C|=3. Find |A?B?C|.

A. 54

Q21. If a set S has 4 elements, what is the cardinality of its power set P(S)?

A. 16

Q22. Number of binary relations from a set A with 3 elements to a set B with 4 elements is?

A. 2^12 =

Q23. Number of all functions from a set A of size 2 to a set B of size 5 is?

A. 5^2 = 2

Q24. Number of equivalence relations on an 3-element set is?

A. 8

Q25. Number of bijective functions (permutations) on an 6-element set is?

A. 6^6 = 4

Q26. If a set S has 5 elements, what is the cardinality of its power set P(S)?

A. 25

Q27. Number of binary relations from a set A with 4 elements to a set B with 2 elements is?

A. 4^2 = 1

Q28. Number of all functions from a set A of size 4 to a set B of size 4 is?

A. 16

Q29. Number of equivalence relations on an 3-element set is?

A. 6

Q30. Number of bijective functions (permutations) on an 3-element set is?

A. 8

Q31. By the pigeonhole principle, among 50 integers, at least how many have the same remainder modulo 7?

A. 8

Q32. In how many ways can 3 distinct prizes be distributed among 6 students if each student can receive any number of prizes?

A. P(6,3) =

Q33. The value of C(11,2) is?

A. 53

Q34. The value of P(9,2) (permutations without repetition) is?

A. 81

Q35. By the pigeonhole principle, among 26 integers, at least how many have the same remainder modulo 5?

A. 2

Q36. In how many ways can 4 distinct prizes be distributed among 6 students if each student can receive any number of prizes?

A. P(6,4) =

Q37. The value of C(15,3) is?

A. 2730

Q38. The value of P(11,4) (permutations without repetition) is?

A. 7924

Q39. By the pigeonhole principle, among 50 integers, at least how many have the same remainder modulo 7?

A. 9

Q40. In how many ways can 4 distinct prizes be distributed among 11 students if each student can receive any number of prizes?

A. P(11,4)

Q41. A fair coin is tossed 3 times. Probability of exactly 2 heads is?

A. 1/4

Q42. Two dice are thrown. Probability that sum is 9 is?

A. 1/9

Q43. A card is drawn from a deck. Probability it is an ace is?

A. 1/13

Q44. If P(A)=0.4, P(B)=0.5 and A,B independent, P(A?B) is?

A. 0.10

Q45. Entropy (in bits) of a binary source with P(0)=P(1)=1/2 is?

A. 2 bits

Q46. Entropy of a certain event (p=1) is?

A. Infinity

Q47. If H(X)=2 bits, it means on average how many yes/no questions are needed?

A. 0.5

Q48. A relation R is a partial order if it is?

A. Reflexi

Q49. An equivalence relation must be?

A. Antisyn

Q50. If f: A?B is both injective and surjective, it is?

A. Only su

Q51. How many onto (surjective) functions exist from a 5-element set to a 2-element set?

A. 30

Q52. In how many ways can the letters of the word 'COMPUTER' (8 distinct letters) be arranged?

A. 2^8 = 2

Q53. For a binary memoryless source with probabilities (1/8, 7/8), the information content of the less likely symbol is greater than?

A. 1 bit

Q54. How many onto (surjective) functions exist from a 4-element set to a 2-element set?

A. 14

Q55. In how many ways can the letters of the word 'COMPUTER' (8 distinct letters) be arranged?

A. 8^8

Q56. For a binary memoryless source with probabilities (1/8, 7/8), the information content of the less likely symbol is greater than?

A. Exactly

Q57. How many onto (surjective) functions exist from a 4-element set to a 2-element set?

A. 16

Q58. In how many ways can the letters of the word 'COMPUTER' (8 distinct letters) be arranged?

A. 2^8 = 2

Q59. For a binary memoryless source with probabilities (1/8, 7/8), the information content of the less likely symbol is greater than?

A. 0 bit

Q60. How many onto (surjective) functions exist from a 7-element set to a 2-element set?

A. 126

Q61. Mutual information I(X;Y) ? 0 with equality iff?

A. X and Y

Q62. For independent X,Y, H(X,Y) equals?

A. max(H(

Q63. Channel capacity is defined as?

A. Maximu

Q64. Huffman coding achieves?

A. Fixed-le

Q65. If A?B then |A?B| equals?

A. |A|

Q66. Principle of inclusion-exclusion for 2 sets is?

A. |A?B|=|

Q67. The number of subsets of size k from n elements is?

A. C(n,k)

Q68. Bayes' theorem relates?

A. Union t

Q69. For a fair 6-faced die, entropy is?

A. 1 bit

Q70. Pigeonhole principle: 13 people imply at least two born in the same?

A. Year

Q71. Number of edges in the complete graph K7 is?

A. 22

Q72. Number of edges in complete bipartite graph K2,2 is?

A. 4

Q73. A tree with 11 vertices has how many edges?

A. 55

Q74. By Cayley's formula, number of labelled spanning trees of K6 is n^(n-2). Its value for n=6 is?

A. 7776

Q75. A connected graph is Eulerian iff?

A. Every v

Q76. A graph has an Euler trail (but not circuit) iff?

A. Exactly

Q77. Maximum number of edges in a simple planar graph with 5 vertices (n?3) is bounded by 3n-6. The bound is?

A. 9

Q78. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K6 or K

Q79. A bipartite graph cannot contain?

A. Isolated

Q80. Handshaking lemma states sum of degrees equals?

A. |E|

Q81. Eccentricity of a vertex v is defined as?

A. max dis

Q82. Radius of a graph is?

A. Maximu

Q83. Diameter of the path graph P6 (n vertices) is?

A. 5

Q84. Centre of a tree consists of?

A. Always

Q85. A full m-ary tree (each internal node has 3 children) of height 2 has at most how many leaves?

A. 6

Q86. A Hamiltonian cycle visits?

A. Only od

Q87. A 4-regular graph on 8 vertices has how many edges?

A. 8

Q88. A rooted tree in which every internal node has at most 2 children is called?

A. Ternary

Q89. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Euleria

Q90. Which statement about trees is FALSE?

A. A tree c

Q91. Number of edges in the complete graph K6 is?

A. 15

Q92. Number of edges in complete bipartite graph K4,4 is?

A. 8

Q93. A tree with 5 vertices has how many edges?

A. 6

Q94. By Cayley's formula, number of labelled spanning trees of K4 is n^(n-2). Its value for n=4 is?

A. 16

Q95. A connected graph is Eulerian iff?

A. Every v

Q96. A graph has an Euler trail (but not circuit) iff?

A. It is dis

Q97. Maximum number of edges in a simple planar graph with 7 vertices (n?3) is bounded by 3n-6. The bound is?

A. 21

Q98. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K6 or K

Q99. A bipartite graph cannot contain?

A. A span

Q100. Handshaking lemma states sum of degrees equals?

A. |V|

Q101. Eccentricity of a vertex v is defined as?

A. max dis

Q102. Radius of a graph is?

A. Maximu

Q103. Diameter of the path graph P8 (n vertices) is?

A. 8

Q104. Centre of a tree consists of?

A. Always

Q105. A full m-ary tree (each internal node has 2 children) of height 3 has at most how many leaves?

A. 5

Q106. A Hamiltonian cycle visits?

A. Every e

Q107. A 5-regular graph on 10 vertices has how many edges?

A. 26

Q108. A rooted tree in which every internal node has at most 2 children is called?

A. Binary

Q109. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Depth-f

Q110. Which statement about trees is FALSE?

A. A tree i

Q111. Number of edges in the complete graph K10 is?

A. 46

Q112. Number of edges in complete bipartite graph K4,2 is?

A. 9

Q113. A tree with 11 vertices has how many edges?

A. 55

Q114. By Cayley's formula, number of labelled spanning trees of K4 is n^(n-2). Its value for n=4 is?

A. 16

Q115. A connected graph is Eulerian iff?

A. Every v

Q116. A graph has an Euler trail (but not circuit) iff?

A. All verti

Q117. Maximum number of edges in a simple planar graph with 5 vertices (n?3) is bounded by 3n-6. The bound is?

A. 10

Q118. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K5 or K

Q119. A bipartite graph cannot contain?

A. An odd

Q120. Handshaking lemma states sum of degrees equals?

A. 2|E|

Q121. Eccentricity of a vertex v is defined as?

A. sum of

Q122. Radius of a graph is?

A. Maximu

Q123. Diameter of the path graph P7 (n vertices) is?

A. 7

Q124. Centre of a tree consists of?

A. Always

Q125. A full m-ary tree (each internal node has 4 children) of height 3 has at most how many leaves?

A. 81

Q126. A Hamiltonian cycle visits?

A. Only od

Q127. A 4-regular graph on 8 vertices has how many edges?

A. 4

Q128. A rooted tree in which every internal node has at most 2 children is called?

A. Bipartit

Q129. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Euleria

Q130. Which statement about trees is FALSE?

A. A tree i

Q131. Number of edges in the complete graph K7 is?

A. 21

Q132. Number of edges in complete bipartite graph K2,2 is?

A. 4

Q133. A tree with 6 vertices has how many edges?

A. 6

Q134. By Cayley's formula, number of labelled spanning trees of K3 is n^(n-2). Its value for n=3 is?

A. 6

Q135. A connected graph is Eulerian iff?

A. Every v

Q136. A graph has an Euler trail (but not circuit) iff?

A. It is dis

Q137. Maximum number of edges in a simple planar graph with 7 vertices (n?3) is bounded by 3n-6. The bound is?

A. 21

Q138. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K4 or K

Q139. A bipartite graph cannot contain?

A. A spann

Q140. Handshaking lemma states sum of degrees equals?

A. |V|

Q141. Eccentricity of a vertex v is defined as?

A. min dis

Q142. Radius of a graph is?

A. Maximu

Q143. Diameter of the path graph P8 (n vertices) is?

A. 8

Q144. Centre of a tree consists of?

A. All leav

Q145. A full m-ary tree (each internal node has 2 children) of height 3 has at most how many leaves?

A. 5

Q146. A Hamiltonian cycle visits?

A. Every v

Q147. A 3-regular graph on 6 vertices has how many edges?

A. 6

Q148. A rooted tree in which every internal node has at most 2 children is called?

A. Spanni

Q149. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Shortes

Q150. Which statement about trees is FALSE?

A. A tree v

Q151. Number of edges in the complete graph K5 is?

A. 11

Q152. Number of edges in complete bipartite graph K3,3 is?

A. 36

Q153. A tree with 7 vertices has how many edges?

A. 21

Q154. By Cayley's formula, number of labelled spanning trees of K4 is n^(n-2). Its value for n=4 is?

A. 64

Q155. A connected graph is Eulerian iff?

A. Every v

Q156. A graph has an Euler trail (but not circuit) iff?

A. Exactly

Q157. Maximum number of edges in a simple planar graph with 4 vertices (n?3) is bounded by 3n-6. The bound is?

A. 8

Q158. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K6 or K

Q159. A bipartite graph cannot contain?

A. Isolated

Q160. Handshaking lemma states sum of degrees equals?

A. |E|

Q161. Eccentricity of a vertex v is defined as?

A. min dis

Q162. Radius of a graph is?

A. Maximu

Q163. Diameter of the path graph P9 (n vertices) is?

A. 4

Q164. Centre of a tree consists of?

A. Always

Q165. A full m-ary tree (each internal node has 4 children) of height 2 has at most how many leaves?

A. 16

Q166. A Hamiltonian cycle visits?

A. Only th

Q167. A 4-regular graph on 8 vertices has how many edges?

A. 4

Q168. A rooted tree in which every internal node has at most 2 children is called?

A. Ternary

Q169. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Shortes

Q170. Which statement about trees is FALSE?

A. A tree c

Q171. In any group, the identity element is?

A. Never u

Q172. Lagrange's theorem states that for finite group G and subgroup H?

A. |H| > |G

Q173. Number of elements in the finite field GF(7^3) is?

A. 343

Q174. Characteristic of a finite field GF(p^n) is?

A. 0

Q175. A linear code with minimum Hamming distance d=5 can correct up to how many errors?

A. 2

Q176. A code with minimum distance d can detect up to how many errors?

A. d-1

Q177. For a Hamming code with 3 parity bits, maximum message length satisfies 2^r ? m+r+1. For r=3, total code length n=2^r-1 equals?

A. 8

Q178. The set of all n-bit vectors under XOR forms?

A. A non-a

Q179. A generator matrix G of a linear (n,k) code has dimensions?

A. k × n

Q180. Cyclic codes are most efficiently encoded/decoded using?

A. Shift re

Q181. In any group, the identity element is?

A. Never u

Q182. Lagrange's theorem states that for finite group G and subgroup H?

A. |H| > |G

Q183. Number of elements in the finite field GF(3^3) is?

A. 6

Q184. Characteristic of a finite field GF(p^n) is?

A. p^n

Q185. A linear code with minimum Hamming distance d=3 can correct up to how many errors?

A. 1

Q186. A code with minimum distance d can detect up to how many errors?

A. d

Q187. For a Hamming code with 4 parity bits, maximum message length satisfies 2^r ? m+r+1. For r=4, total code length n=2^r-1 equals?

A. 16

Q188. The set of all n-bit vectors under XOR forms?

A. Only a

Q189. A generator matrix G of a linear (n,k) code has dimensions?

A. n × n

Q190. Cyclic codes are most efficiently encoded/decoded using?

A. Shift re

Q191. In any group, the identity element is?

A. Always

Q192. Lagrange's theorem states that for finite group G and subgroup H?

A. |H| divi

Q193. Number of elements in the finite field GF(3^2) is?

A. 9

Q194. Characteristic of a finite field GF(p^n) is?

A. n

Q195. A linear code with minimum Hamming distance d=5 can correct up to how many errors?

A. 5

Q196. A code with minimum distance d can detect up to how many errors?

A. d-1

Q197. For a Hamming code with 5 parity bits, maximum message length satisfies 2^r ? m+r+1. For r=5, total code length n=2^r-1 equals?

A. 32

Q198. The set of all n-bit vectors under XOR forms?

A. An abe

Q199. A generator matrix G of a linear (n,k) code has dimensions?

A. k × n

Q200. Cyclic codes are most efficiently encoded/decoded using?

A. K-maps

ANSWER KEY

1-B, 2-A, 3-C, 4-A, 5-A, 6-A, 7-A, 8-A, 9-D, 10-D

11-A, 12-D